شبکه مخابرات داده　　　　تمرین سری 4　　　　فرزان آقایی 401101202

سؤال ۱:

a) final routing tables:

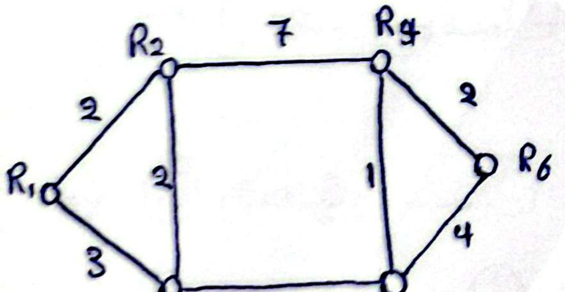
to \ from	R_1	R_2	R_3	R_4	R_5	R_6
R_1	—	5 R_3	3 R_1	12 R_2	12 R_3	16 R_5
R_2	5 R_3	—	2 R_2	7 R_2	8 R_4	12 R_5
R_3	3 R_3	2 R_3	—	9 R_2	9 R_3	13 R_5
R_4	12 R_3	7 R_4	9 R_2	—	1 R_4	5 R_5
R_5	12 R_3	8 R_4	9 R_5	1 R_5	—	4 R_5
R_6	16 R_3	12 R_4	13 R_5	5 R_5	4 R_6	—

← در نتیجه تنها link هایی که استفاده نمی شوند linkهای بین R_1-R_2 و بین R_4-R_6 هستند

↓

2 لینک بی استفاده

b) final routing tables:



to \ from	R_1	R_2	R_3	R_4	R_5	R_6
R_1	—	2 R_1	3 R_1	9 R_2	10 R_4	11 R_4
R_2	2 R_2	—	2 R_2	7 R_2	8 R_4	9 R_4
R_3	3 R_3	2 R_3	—	9 R_2	9 R_3	11 R_4
R_4	9 R_2	7 R_4	9 R_2	—	1 R_4	2 R_4
R_5	10 R_2	8 R_4	9 R_5	1 R_5	—	3 R_4
R_6	11 R_2	9 R_4	11 R_2	2 R_6	3 R_4	—

← در نتیجه تنها link بدون استفاده لینک بین R_5-R_6 است

↓

1 لینک بی استفاده

192.168.8.0/21 → 192.168.8.0 to 192.168.15.255 → int 0   سوال 2 :

192.168.16.0/20 → 192.168.16.0 to 192.168.31.255 → int 1

10.10.0.0/22 → 10.10.0.0 to 10.10.3.255 → int 2

o.w → int 3

a) 192.168.18.45 → interface 1

b) 192.168.14.200 → interface 0

c) 192.168.4.1 → interface 3

d) 10.10.1.5 → interface 2

e) 10.20.5.9 → interface 3

---

a) Ⓐ: 0ms - 3ms → Ⓒ: 3ms - 11ms → Ⓕ: 11ms - 15ms → Ⓗ: 15ms - 21ms   سوال 3 ,
→ Ⓚ: 21ms - 24ms → Ⓑ: 24ms - 27ms → Ⓓ: 27ms - 35ms → Ⓖ: 35ms - 38ms
→ Ⓘ: 38ms - 44ms → Ⓛ: 44ms - 47ms → Ⓔ: 47ms - 50ms → Ⓙ: 50ms - 56ms → $\boxed{56\text{ms}}$

b) Ⓐ, 0ms - 3ms → Ⓒ: 3ms - 11ms → Ⓓ, 11ms - 19ms → Ⓕ, 19ms - 23ms
→ Ⓗ, 23ms - 29ms → Ⓘ, 29ms - 35ms → Ⓙ, 35ms - 41ms → Ⓚ, 41ms → 44ms
→ Ⓑ, 44ms → 47ms → Ⓔ, 47ms - 50ms → Ⓖ, 50ms - 53ms → Ⓛ, 53ms - 56ms → $\boxed{56\text{ms}}$

سوال 3،

c) deficit → [illegible]ms ⇒ ~~[illegible]~~ Ⓗ, 15ms →

Ⓐ, ~~[illegible]~~ 17ms → Ⓚ, 24ms → Ⓒ, $2\overset{7}{8}$ ms →

Ⓕ, 28ms → Ⓘ, 31ms → Ⓑ, 41ms →

Ⓙ, 47ms → Ⓛ, 48ms → Ⓖ, 51ms →

Ⓓ, 53ms → Ⓔ, 56ms → $\boxed{56\text{ms}}$

سوال 4،

a) 56 bytes each 4μs → 112×10^6 bits each 1s → $\boxed{112\text{ Mbps}}$

b) 56 bytes per 3μs → $\boxed{149.33\text{ Mbps}}$

سوال 5،

a) at P1 : input : in_if = 1 in_lbl = 0 $\xrightarrow{\text{LFIB}}$ output : out_if = 2 out_lbl = 12

→ label chaaged to 12 → P2

at P2 : input : in_if = 1 in_lbl = 12 $\xrightarrow{\text{LFIB}}$ out_if = 2 out_lbl = 0

→ Label changed to 0 → PE_out1 → label is removed

⇒ P1 → P2 → PE_out1

lbl0 → lbl12　lbl12 → lbl0　remove lbl

b) at P2 , input , in_if = 2 in_lbl = 0 $\xrightarrow{LFIB}$ output , if = 3 lbl = 41 سوال 5.

→ Label changed to 41 → P4

at P4 , in_if = 2 in_lbl = 41 $\xrightarrow{LFIB}$ out_if = 1 out_lbl = 0

→ Label changed to 0 → PE-in2 → Label is removed

⇒ P2 → P4 → PE-in2
0 → 41 41 → 0 0 removed

c) at P4 , in_if = 1 in_lbl = 0 $\xrightarrow{LFIB}$ out_if = 3 out_lbl = 50

→ Label changed to 50 → P3

at P3 , in_if = 2 in_lbl = 50 $\xrightarrow{LFIB}$ out_if = 3 out_lbl = 0

→ label changed to 0 → PE-out2 → Label removed

⇒ P4 → P3 → PE-out2
0 → 50 50 → 0 Label remove

سوال 6.

a) 200.100.0.0/21 → 200.100.0.0 to 200.100.7.255

A → 300 ⇒ 512 ⇒ from 200.100.0.0 to 200.100.1.255

B → 500 ⇒ 512 ⇒ from 200.100.2.0 to 200.100.3.255

C → 100 ⇒ 128 ⇒ from 200.100.4.0 to 200.100.4.127

D → 50 ⇒ 64 ⇒ from 200.100.4.128 to 200.100.4.191

E → 200 ⇒ 256 ⇒ from 200.100.5.0 to 200.100.5.255

b) total addresses → $2^{11} = 2048$ used addresses → $512+512+256+128+64 = 1472$

→ remaining addresses $= 2048 - 1472 = \boxed{576}$

c) A → 300 from 512 → %41.41 wasted

B → 500 from 512 → %2.34 wasted

C → 100 from 128 → %21.88 wasted

D → 50 from 64 → %21.88 wasted

E → 200 from 256 → %21.88 wasted

$\boxed{A \rightarrow \%41.41}$